Multi-Step Successor-Measure TD for Play Data: Dwell Structure as the Approximation Scheme

Working note — Factored-FB project
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Abstract

Successor-measure methods (FB and relatives) learn the discounted state-occupancy of a policy by one-step temporal-difference (TD) learning. On manipulation play data, occupancy over goal-relevant object coordinates is dominated by dwell: objects rest at stable configurations for tens to hundreds of steps between brief, action-driven transports. We show that one-step TD is structurally mismatched to this regime—its fixed point under sparse anchoring degenerates to the action-independent dwell constant—and derive a multi-step (k-step / λ-return) measure-TD operator whose error decomposition is gated by the data's interaction sparsity: the off-policy bias of the observed k-step window is bounded by $\gamma^{T_{\text{int}}}$, where T_{int} is the time to the first interaction with the tracked object. Play data thus licenses long observed returns precisely where one-step bootstrapping fails. The dwell scale sets the lower edge of the useful horizon and the discount half-life the upper; a geometric horizon recovers the λ-return with a strictly better contraction modulus and no dataset-specific constant. The data are untouched: the structure enters through the estimator only.

1 Setup

Let $(\mathcal{S}, \mathcal{A}, P, \gamma)$ be an MDP with $\gamma \in (0,1)$. A known projection $c : \mathcal{S} \to \mathbb{R}^{d_c}$ extracts the goal-relevant object coordinate (e.g. cube position). For a goal g and radius ε, write $X = B_\varepsilon(g)$ and $h(s) = \mathbf{1}[c(s) \in X]$. Data are collected by a behavior policy β; we seek the successor measure of a target policy π,

$$M^\pi(s, a, X) \;=\; \mathbb{E}^\pi\Big[\textstyle\sum_{t=1}^{\infty} \gamma^{\,t-1} h(s_t) \;\Big|\; s_0 = s,\, a_0 = a\Big], \tag{1}$$

counting from the next state and bounded by $(1-\gamma)^{-1}$. Evaluating M^π against a reward r supported on object coordinates yields $Q_r^\pi(s,a) = \int M^\pi(s,a,\mathrm{d}x)\, r(x)$, the identity underlying zero-shot readout in FB-style methods.

Empirical regime (measured, not assumed). Across eleven manipulation datasets (OGBench cube single/double/triple/quadruple and scene; Metaworld play-resample in two regimes and two tasks; two expert sets), region occupancy factors as mass $\approx P(\text{arrive}) \cdot \mathbb{E}[\gamma^{T_{\text{arr}}}] \cdot \frac{1-\gamma^L}{1-\gamma}$ with log-correlation ≥ 0.91, where L is the in-region sojourn length; the single longest sojourn carries 74–99% of in-region time; and mean sojourns grow from 22 steps (cube-single) to 140 steps (cube-quadruple). Object coordinates are exactly stationary on 48–92% of transitions. We refer to this two-timescale, piecewise-constant structure as dwell structure.

2 The k-step Bellman operator for measures

Splitting the sum in equation 1 at time $k \geq 1$,

$$\sum_{t=1}^{\infty} \gamma^{t-1} h(s_t) = \underbrace{\sum_{t=1}^{k} \gamma^{t-1} h(s_t)}_{\text{observed}} + \gamma^k \sum_{u=1}^{\infty} \gamma^{u-1} h(s_{k+u}), \tag{2}$$

and taking expectations under π gives, for every k,

$$M^\pi(s,a,X) = (\mathcal{T}_k M^\pi)(s,a,X) \;:=\; \mathbb{E}^\pi\Big[\sum_{t=1}^{k} \gamma^{t-1} h(s_t)\Big] + \gamma^k\, \mathbb{E}^\pi\big[M^\pi(s_k,a_k,X)\big]. \quad (3)$$

Proposition 1 (Contraction and fixed point). For any bounded M, M', $\|\mathcal{T}_k M - \mathcal{T}_k M'\|_\infty \le \gamma^k \|M - M'\|_\infty$. Hence $\mathcal{T}_k$ is a γ^k-contraction whose unique fixed point is M^π for every $k \ge 1$.

Proof. The observed term is M-independent; the bootstrap terms differ by $\gamma^k \mathbb{E}^\pi[M - M'] \le \gamma^k \|M - M'\|_\infty$. Uniqueness by Banach; equation 3 shows M^π is a fixed point. □

The choice of k therefore never changes the target, only the error dynamics: k=1 is standard measure TD, $k \to \infty$ is Monte-Carlo occupancy counting.

3 Error decomposition and the off-policy bias

Write a learned measure as $\widehat{M} = M^\pi + e$. Applying the empirical operator—trajectory actions from β, bootstrap action from π—the post-update error decomposes as

$$e' \;=\; \underbrace{\gamma^k\, \mathbb{E}\big[e(s_k,a_k)\big]}_{\text{bootstrap error, } \gamma^k\text{-shrunk}} \;+\; \underbrace{\Delta_k^{\beta\pi}}_{\text{off-policy bias}} \;+ \text{sampling noise}, \quad (4)$$

$$\Delta_k^{\beta\pi} \;=\; \mathbb{E}^\beta\Big[\sum_{t=1}^{k} \gamma^{t-1} h(s_t)\Big] - \mathbb{E}^\pi\Big[\sum_{t=1}^{k} \gamma^{t-1} h(s_t)\Big].$$

$\Delta_1^{\beta\pi} = 0$ since the first transition is conditioned on (s,a); generically $|\Delta_k|$ grows with k—the classical argument for small k. Dwell structure breaks that argument.

Assumption 1 (Interaction-gated object dynamics). There is an interaction indicator $\iota(s,a) \in \{0,1\}$ (contact with the tracked object) such that $c(s_{t+1}) = c(s_t)$ whenever $\iota(s_t,a_t) = 0$.

This is the measured piecewise-constancy above (exact to numerical precision in the simulators considered). Let $T_{\text{int}} = \min\{t \ge 1 : \iota(s_t,a_t) = 1\}$ be the first interaction time after (s_0,a_0).

Lemma 1 (Dwell invariance). On the event $\{k \le T_{\text{int}}\}$, the observed sum is policy-independent: $\sum_{t=1}^{k} \gamma^{t-1} h(s_t) = h(s_1)\,\frac{1-\gamma^k}{1-\gamma}$, a function of the initial object position only. Consequently

$$\big|\Delta_k^{\beta\pi}\big| \;\le\; \mathbb{E}^\beta\Big[\gamma^{\,T_{\text{int}}}\Big] \cdot \frac{1 - \gamma^{\,(k - T_{\text{int}})^+}}{1-\gamma}. \quad (5)$$

Proof. While $t < T_{\text{int}}$, Assumption 1 freezes $c(s_t) = c(s_1)$, hence $h(s_t) = h(s_1)$ regardless of the actions taken; the partial sum is the stated geometric series under every policy. Policies can disagree only at $t > T_{\text{int}}$; bounding the disagreement of the remaining $(k - T_{\text{int}})^+$ terms by the maximal indicator sum and discounting by $\gamma^{T_{\text{int}}}$ yields equation 5. □

Remark 1 (Where the bound bites). The bias is gated by $\mathbb{E}[\gamma^{T_{\text{int}}}]$: the discount decays the window before the policies can disagree about the object. Measured interaction gaps give $\gamma^{T_{\text{int}}} \approx 0.4$–$0.55$ on cube-single ($T_{\text{int}} \sim 60$–90) and ≈ 0.17 on cube-double ($T_{\text{int}} \sim 176$): the safe-horizon budget is largest on the multi-object environments where one-step TD fails hardest. In the opposite regime (navigation: the agent is the object, $\iota \equiv 1$, $T_{\text{int}} = 1$) the bound is vacuous and small-k orthodoxy applies. The admissible horizon is itself a function of the data's dwell structure.

4 Choosing the horizon

Lower edge: the dwell scale. For a rest region with sojourn length L and arrival time T_{arr}, the observed part of the target carries $\mathbb{E}[G^{(k)}_{\text{obs}}] \approx P(T_{\text{arr}} \le k) \cdot \frac{1-\gamma^{\min(L,\, k-T_{\text{arr}})}}{1-\gamma}$ while the bootstrap carries weight γ^k. For $k \ll L$ the dwell mass $\frac{1-\gamma^L}{1-\gamma}$ ($\approx$43 at L=56, γ=0.99) sits inside the bootstrap—the quantity TD must reconstruct by iterating on itself. At k=1 the observed signal fires at the one-step hit rate (0.05% per sample on cube-double regions). For $k \gtrsim L$ the full sojourn mass moves into the observed, policy-invariant part. Hence k should exceed the median sojourn (22–60 steps on our datasets).

Upper edge: the discount half-life. Past $k^* = \ln 2/\ln(1/\gamma) \approx 69$ steps (γ=0.99), γ^k has halved and further observed steps contribute $O(\gamma^k)$ mass while adding variance; the useful range is $k \in [L_{\text{med}}, \sim 2k^*] \approx [50, 150]$, fixed by two measurable quantities rather than tuned.

No fixed k: the λ-return. Drawing $k \sim \text{Geom}(1-\lambda)$ per update yields the expected operator

$$\mathcal{T}_\lambda = (1-\lambda) \sum_{n \ge 1} \lambda^{\,n-1}\, \mathcal{T}_n, \tag{6}$$

the classical λ-return for measures.

Proposition 2 (Modulus of $\mathcal{T}_\lambda$). $\mathcal{T}_\lambda$ is a contraction with modulus $(1-\lambda)\sum_{n\ge1}\lambda^{n-1}\gamma^n = \frac{\gamma(1-\lambda)}{1-\lambda\gamma} < \gamma$ for every $\lambda \in (0,1)$, with on-policy fixed point M^π for all λ.

Proof. Mix Proposition 1 over the geometric weights; the series sums to the stated modulus. The fixed point is shared by every $\mathcal{T}_n$. □

Off-policy without corrections, the fixed point of the mixed empirical operator interpolates between M^β ($\lambda \to 1$) and the one-step off-policy solution ($\lambda \to 0$); Lemma 1 applies to each $\mathcal{T}_n$ inside the mixture, so the interpolation error is again gated by $\gamma^{T_{\text{int}}}$. Setting the mean horizon $(1-\lambda)^{-1} \approx (1-\gamma)^{-1}$ matches the mixture to the discount and removes the per-dataset constant. Truncation at an episode boundary replaces the tail with the γ^k-weighted bootstrap and preserves the operator.

5 Contrast: the one-step collapse

At k=1, $\mathcal{T}_1 M = \mathbb{E}[h(s_1)] + \gamma\mathbb{E}[M(s_1,a_1)]$. For a sparse region, $\mathbb{E}[h(s_1)] \approx 0$ on almost every sample, so the empirical operator is approximately $\gamma\times$(a projection of M's own average). Any near-constant function C satisfies $\mathcal{T}_1 C \approx \bar{p} + \gamma C$ with $\bar{p}$ the dataset-average one-step hit rate, whose stable point $C^* = \bar{p}/(1-\gamma)$ is the dataset-average mass of the region—the dwell-implied constant. This is the degenerate fixed point we observe empirically on cube-double (measured mass 43.5 vs. dwell-implied 43.0, with no state–action variation): starved of anchors, one-step TD converges to the mass level and loses the ranking. Under $\mathcal{T}_k$ with $k \gtrsim L$, the anchor probability per update grows from P(hit in one step) to P(hit within k), and the constant arrives pre-assembled in the observed term; the degenerate fixed point loses its basin.

6 Estimator, not data

No preprocessing is performed: no event detection, no trajectory segmentation, no relabeling into aligned sub-demonstrations. The sampler reads raw length-k windows from stored trajectories (the same access pattern used by hindsight goal sampling), and the structure enters exclusively through (i) the multi-step operator above, and (ii) optionally a structured measure head $M(s,a,g) = D(g)\cdot H(s,a,g)$ with $D \ge 0$ a goal-only magnitude and $H \in [0,1]$ a bounded bilinear factor $\phi(s,a)^\top\psi(g)$, which gives the action-independent dwell magnitude a dedicated low-capacity container (a dueling-style decomposition for measures). The zero-shot readout $z_r = \mathbb{E}_\rho[\psi(g)D(g)r(g)]$, $Q_r = \phi^\top z_r$ is unchanged from FB.

Pre-registered empirical gates (to be run)

(1) On cube-double phase-2 conditions, the learned mass must not pin to the dwell constant, and signed state–action variation must survive. (2) Performance should be flat in k once $k \gtrsim L_{\text{med}}$ and degrade only as $k \to 1$; sharp tuning in k falsifies the mechanism as stated. (3) Calibration is scored against the λ-return fixed point, not the one-step one, when $\lambda > 0$.

References